The paradox of reversible processes and the local thermodynamic interpretation of general relativity

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Abstract

Classical thermodynamics defines entropy through reversible processes, yet a reversible process, which consists of transitions between equilibrium states, looks contradictory. Equilibrium states do not change, but a process requires a departure from equilibrium, since something has to drive the transitions. We review the main contemporary responses to this paradox: Norton's account of reversible processes as approximations rather than idealizations, the quasi-process view that Valente and Uffink trace to Ehrenfest-Afanassjewa, epistemic readings, and axiomatic reconstructions that dispense with processes altogether. We also examine the dual time-scale argument, which separates a fast relaxation time from a slow driving time, and conclude that it clarifies some aspects of the paradox but does not remove it. Jacobson's derivation of the Einstein field equations of general relativity (GR) from the Clausius relation on local Rindler horizons is then considered in both directions: what the derivation says about the paradox of locally reversible processes, and what the paradox says about the limitations of GR if the locally reversible thermodynamic interpretation of GR is valid.

1 Introduction

Clausius defined the change in entropy between two equilibrium states as $\int đQ_{\text{rev}}/T$ along a reversible process connecting them. Textbooks describe such a process as one that passes only through equilibrium states and that can be run backward by an infinitesimal change in the external conditions [1, p. 4], [2, p. 22], [3, pp. 191–192]. Norton [4, p. 43] pointed out that this description cannot be taken literally. An equilibrium state does not change unless it is disturbed, so a process consisting solely of equilibrium states cannot have transitions. Norton calls this the paradox of thermodynamically reversible processes, and his survey of the literature since Carnot finds the same tension in nearly every textbook characterization [4, §7]. His resolution that reversible processes are approximations to real irreversible processes and not idealized processes has prompted a noticeable literature [5, 6, 7].

The paradox matters for gravity as well. Since Jacobson [8] derived Einstein's equation from the Clausius relation $\delta Q = T\,dS$ applied to local Rindler horizons, many have read general relativity (GR) as an equation of state that holds when spacetime is in local equilibrium [9, 10, 11]. If the Clausius relation for real thermodynamic processes is only approximate, we can ask whether Einstein's equation suffers the same fate.

Section 2 states the paradox. Section 3 reviews the main contemporary positions. It also discusses the dual time-scale argument, which is the response physicists usually give first, and examines its limitations. Section 4 turns to Jacobson's derivation. There we ask why the derivation succeeds even though reversible processes are problematic. Section 5 argues that, on the local thermodynamic interpretation, GR is unlikely to be the full effective theory of gravity and explains why black holes present so many problems for semiclassical gravity. Section 6 concludes.

2 The paradox

Norton states the paradox as a conflict between two requirements [4, p. 43]. Splitting each requirement in two gives the four claims that the textbook characterization of a reversible process makes [1, p. 4], [2, p. 22]: (i) a reversible process is a process, so the state of the system changes in time; (ii) it passes only through equilibrium states; (iii) an equilibrium state does not change unless it is disturbed; and (iv) change requires an imbalance, such as a difference of pressure, temperature or chemical potential. Claims (ii) and (iii) imply that nothing happens, and claims (i) and (iv) imply that something is out of balance. The four claims cannot all be true.

Technically, thermodynamics as a theory of equilibrium states is not threatened by the paradox. The state functions U, S and F are defined on equilibrium state space, and their differences do not depend on the path, possibly through non-equilibrium states, that a real process takes from one equilibrium to another. The trouble lies in how the curves of equilibrium

states labelled ‘reversible processes’ are interpreted as relating to real processes. For example, can we really guarantee that real processes remain at least somewhat close to reversible processes at all times? Are Carnot engines realizable in an ideal environment? The paradox suggests negative answers to these questions, in contrast to textbook treatments.

3 Contemporary positions

3.1 Norton: approximation, not idealization

Norton [12], [4, §3.3] distinguishes an idealization, a real or fictitious system whose properties give an inexact description of a target system, from an approximation, an inexact description of the target that does not refer to any other system. A frictionless plane is an idealization, since a plane without friction can be described consistently. A reversible process cannot be an idealization, since there is no consistent description satisfying the four claims in Section 2. Norton concludes that the term ‘thermodynamically reversible process’ names a set of real irreversible processes, and that the properties attributed to a reversible process are limit properties of that set as the driving forces go to zero [4, §2.2, p. 45]. The limiting states form a curve of equilibrium states, a set of states with no time ordering. Quantities computed on the curve, such as $\int đQ/T$, approximate the corresponding quantities of real processes, but the curve itself does not describe any process. Norton notes that Duhem had given essentially the same account in 1903 [13, pp. 59–74]. Callen’s textbook makes a related point. It defines a quasi-static process as a dense succession of equilibrium states and stresses that it is an idealized concept, distinct from any real process because it involves no rates or time [14, p. 96]. Callen then relates it to real processes that pass through a sequence of equilibrium states with brief departures between them [14, pp. 96–97], which is the picture of iterated equilibria discussed in the next subsection.

Norton adds a second point that will matter in Section 5. At molecular scales the imbalance that drives a process must exceed thermal fluctuations if the process is to proceed reliably in one direction, and this costs a minimum amount of entropy [15, 7], [4, §5]. There the reversible limit cannot even be approached. Norton uses this fluctuation no-go result against analyses of Maxwell’s demon and of Landauer’s principle that rely on reversible molecular steps [16, 17]. More generally, whether a reversible curve is a good approximation depends on how large the system is compared with its fluctuations.

3.2 Quasi-processes and quasi-static processes

Ehrenfest-Afanassjewa [18, pp. 11–13] distinguished quasi-processes from quasi-static processes. A quasi-process is a continuous curve of equilibrium states, which she treated as a mathematical object without direction and deliberately did not call reversible. A quasi-static

process is a real process that passes through a dense discrete sequence of equilibrium states, with near-equilibrium passages between them. In her account 'infinitely slow' is not meant literally. A quasi-static process is close to a quasi-process when the deviations are below what the available instruments can detect. Valente [5] and Uffink and Valente [6] argue that this distinction anticipates and resolves Norton's paradox.

Norton discusses this view and is not persuaded [4, §7.11, pp. 59–60]. The states at which a quasi-static process pauses are not exact equilibrium states. They count as equilibria only because their departure from equilibrium is too small to measure. Moreover, in generic systems the relaxation to one of these quasi-equilibria is asymptotic and never completes. While measurement error can hide the difference between the quasi-static process and the quasi-process at any single step, it cannot hide the accumulated difference over many steps. This accumulation problem returns in Section 5.3.

This suggests that the paradox is really about finite-time approachability. Taken as a quasi-process, a curve of equilibrium states is perfectly consistent, because it says nothing about how the system gets from one state to the next. Problems appear when we want the curve to describe something that actually happens.

3.3 Epistemic views

On Jaynes's view [19], an equilibrium state is the maximum-entropy distribution compatible with the macroscopic constraints. In this epistemic (agent-relative) view, a reversible process is one in which transitions between the maximum-entropy distributions involve no loss of information. This can only work in the slow-process limit in which some quantities are adiabatic invariants.

This epistemic view has received the expected criticisms. Albert [20, p. 64], for example, asks whether anyone can seriously believe that our ignorance of the exact microstate is what makes milk dissolve in coffee as an irreversible process. The milk spreads the same way regardless of whether anyone knows its microstate, so irreversibility here seems to be ontic rather than epistemic. Norton's critique of information-theoretic thermodynamics [16, 17] adds that any physical agent who would use the information is also subject to the fluctuation floor.

3.4 Defining entropy without reversible processes

While reversible processes have often been used to define entropy, it is possible to define entropy purely from the differential forms of the laws of thermodynamics. On equilibrium state space the first law gives the one-form $\omega = dU + P\,dV$, which involves only state functions. Carathéodory's form of the second law then leads to $\omega = T\,dS$ [21, 22]. Entropy differences become integrals along curves in state space rather than along processes.

Lieb and Yngvason [23] go further and derive entropy from an order relation of adiabatic accessibility, without assuming that state space has a differentiable structure. Reversibility survives only as mutual accessibility, $A \sim B$, which Lieb and Yngvason concede can be realized

only in an idealized sense that would take infinitely long [23, p. 17], [4, §7.5]. Brown and Uffink [24] state the assumption that an isolated system approaches a unique equilibrium state as a separate 'minus first law', which implies that a reversible process, even as an approximation, is not granted and has to be postulated.

3.5 Fast time and slow time: the dual time-scale argument

Physicists usually answer the paradox by appeal to time scales. Let τ_{relax} be the time a system needs to relax to equilibrium, τ_{proc} the time over which the external conditions change, and $\varepsilon = \tau_{\text{relax}}/\tau_{\text{proc}}$. A multiple-scale expansion [25] writes the state as

$$x(t) = x_0(s) + \varepsilon\, x_1(t, s) + O(\varepsilon^2), \qquad s = \varepsilon t, \tag{1}$$

with a fast time t and a slow time s. On the fast scale x_0 is a fixed point, the equilibrium state for the constraints at slow time s. On the slow scale $x_0(s)$ drifts. According to the argument, the paradox confuses two derivatives: 'equilibrium' refers to $\partial_t x_0 = 0$ and 'process' to $dx_0/ds \neq 0$, and both can hold at once. Local equilibrium thermodynamics [26] has the same structure. Each fluid cell is in equilibrium on the scale of molecular collisions while its state variables vary on the scale of the flow.

The argument is useful, but it has three limitations. First, it does not address Norton's paradox, since slowness is not reversibility. Real processes have a finite ε, and for them the drift of x_0 is driven by the $O(\varepsilon)$ correction x_1, a genuine departure from equilibrium that produces entropy of order ε. Such processes are real, but there is no guarantee of reversibility. The limit $\varepsilon \to 0$ is well defined, but what it yields is the curve $x_0(s)$ of equilibrium states, traversed infinitely slowly in physical time. Related to this, Uffink [27, p. 344] and Norton [4, §§7.5, 7.9] stress that quasi-static and reversible are different notions. Heat conducted slowly through a thin wire between two reservoirs at fixed temperatures is quasi-static for both reservoirs and yet irreversible, and so is gas effusing slowly through a pinhole. Second, the argument requires a kinetic foundation from outside equilibrium thermodynamics, implying that equilibrium thermodynamics is not self-sufficient and is incomplete. The ratio ε is fixed by kinetics, while the theory itself contains no time scales. Finally, the separation of time scales can fail. Relaxation slows down near critical points, glasses and systems with hysteresis do not relax on any practical time scale, and at molecular scales fluctuations blur the distinction between fast and slow variables.

In short, while the dual time-scale argument shows how a reversible process may mathematically be defined consistently, it does not show whether such a process can even ideally exist.

4 General relativity as local equilibrium

4.1 Jacobson's derivation

For a congruence of null geodesics with tangent k^a and affine parameter λ, the expansion θ, shear σ_{ab} and twist ω_{ab} obey the Raychaudhuri equation, which in four dimensions reads

$$\frac{d\theta}{d\lambda} = -\tfrac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}k^a k^b. \tag{2}$$

The equation holds for any Lorentzian metric with the Levi-Civita connection, and no field equation is assumed.

Jacobson's derivation [8] goes as follows. At a point p and for a null direction k^a, choose a small spacelike two-surface through p such that the orthogonal null congruence has $\theta = \sigma = 0$ at p. The past part of this congruence forms a local Rindler horizon. Since its expansion and shear vanish at p, we say that the horizon is in equilibrium at p. Near p there is an approximate boost Killing vector $\chi^a \approx -\kappa\lambda k^a$, and accelerated observers who stay outside the horizon see the vacuum at the Unruh temperature $T = \hbar\kappa/2\pi$. The energy flux across the horizon, measured with respect to χ^a, is the heat

$$\delta Q = -\kappa \int \lambda\, T_{ab} k^a k^b \, d\lambda \, dA, \tag{3}$$

where T_{ab} is the stress-energy tensor of matter. With $\theta(p) = \sigma(p) = 0$, Equation (2) gives $\theta \approx -\lambda R_{ab} k^a k^b$ to leading order, so the area changes by

$$\delta A \approx -\int \lambda R_{ab} k^a k^b \, d\lambda \, dA. \tag{4}$$

Jacobson then postulates that entropy is proportional to area, $S = \eta A$, and imposes the Clausius relation $\delta Q = T\,\delta S$ for all null k^a. This gives $R_{ab} + \Phi g_{ab} = (2\pi/\hbar\eta)T_{ab}$ for some function Φ. Conservation of T_{ab} and the contracted Bianchi identity then fix Einstein's equation with a cosmological constant and $G = 1/(4\hbar\eta)$.

What picks out GR in the derivation is the area-entropy relation. If the entropy is a different functional of the geometry, the Clausius relation gives different field equations, as in $f(R)$ gravity, where the Wald entropy [28] is proportional to $f'(R)A$ [11].

4.2 Why a reversible process matters for the Jacobson derivation of GR

The Jacobson derivation relies on the Clausius equality $\delta Q = T\,dS$, which holds only for reversible processes. Therefore, since this relation is imposed at every point and for every null direction, the derivation commits to local thermodynamic reversibility, including where the global process is irreversible. A locally reversible process is still a reversible process, and

is subject to the same constraints as any reversible process. Hence the paradox of Section 2 applies to GR in the local thermodynamic interpretation.

One could reply that only the relation $\delta Q = T\,dS$ matters, and that the physical story behind the relation can be dropped. On this reading the Clausius relation is a constraint between state functions on equilibrium horizons, in the spirit of the axiomatic reconstructions of Section 3, and the paradox never arises. That reply is consistent, but it gives up most of what made the derivation interesting, because Jacobson's argument was meant to explain Einstein's equation as the equation of state of a system with microscopic degrees of freedom.

4.3 Equilibrium at a point and a family of local Rindler horizons

In the derivation, equilibrium holds only at the point p. Consider the congruence built at p, and suppose that $R_{ab}k^a k^b > 0$. Integrating $\theta \approx -\lambda R_{ab}k^a k^b$ near p gives

$$A(\lambda) \approx A(p)\left[1 - \tfrac{1}{2}R_{ab}k^a k^b\,\lambda^2\right] \tag{5}$$

for the cross-sectional area of the horizon. The area therefore has p as a turning point, since the expansion changes from positive to negative at p. The horizon is thus stationary at p and changing everywhere else along the congruence.

Since expansion and shear are properties of a congruence rather than of a single geodesic, this equilibrium belongs to the chosen horizon and not to the geodesics that generate it. The same null geodesic can serve as a generator of a local horizon at another point q, but only if a different two-surface is chosen there, one with $\theta(q) = \sigma(q) = 0$. The derivation therefore relies on a whole family of local Rindler horizons, each in equilibrium at a single point. Concretely, for each point p of spacetime and each null direction k^a at p, the family contains one local Rindler horizon whose expansion and shear vanish at p, and that horizon is in equilibrium only there. No horizon in this family is ever required to pass from one equilibrium to the next, so the derivation never asks a single horizon to stay in equilibrium along an extended path.

4.4 Missing pieces of dynamics in the Jacobson derivation

The derivation is perturbative about p, with the only dynamical law used along the affine parameter being the Raychaudhuri equation, and only in its linearized form $\theta \approx -\lambda R_{ab}k^a k^b$. The factor λ in the heat integral comes from the boost vector and is kinematic once χ^a is fixed.

Three pieces of dynamics are left out. The first is the quadratic part of the Raychaudhuri equation. With $\theta(p) = \sigma(p) = 0$, the shear grows as $\sigma \approx -\lambda C$, where C stands for the Weyl component $C_{acbd}k^c k^d$, so θ^2 and σ^2 are $O(\lambda^2)$ in $d\theta/d\lambda$ and change the area only at $O(\lambda^4)$, while the heat and the leading area change are $O(\lambda^2)$. The second is relaxation. In ordinary thermodynamics a system reaches equilibrium by relaxation, which is asymptotic (Section 3).

In Jacobson's derivation nothing relaxes. The third is that the boost is a Killing vector only to leading order, so the temperature and the local vacuum are also defined only to that order.

This is why GR can be derived even though reversible processes are problematic: the dynamics that causes the problem, relaxation and dissipation, is either stipulated away or pushed to higher order. When these pieces are restored, the problem returns. Over a finite stretch of generators the expansion is of order $\lambda R_{ab}k^a k^b$ everywhere except at p, so the horizon is out of equilibrium along the whole stretch, and θ plays the role of the driving imbalance. The quadratic terms then change the area by an amount with no matching heat. Its relative size is of order $(\ell/L)^2$ for a stretch of length ℓ and curvature radius L, and it is quadratic in the imbalance, as entropy production near equilibrium generally is [26]. The reversible Clausius equality is recovered only as the stretch shrinks to the point p.

5 Why GR may not be the full effective theory

By an effective theory of gravity we mean a description that captures everything relevant about gravitational dynamics in a coarse-grained regime, typically the regime of low energy and low curvature. GR is usually taken to be that theory wherever curvature is small [29, 30]. In this section, we argue that there is reason to doubt GR as the full effective theory of gravity, though we also note that GR remains valid for ordinary phenomena and most astronomical phenomena, with the notable exception of black holes. The local thermodynamic interpretation indicates what the missing pieces would look like.

5.1 Finite-time approachability at every point

Jacobson's derivation of GR is mathematically consistent, since it imposes equilibrium only at single points. However, it faces the finite-time approachability problem of Section 3. A local horizon patch has finite size, so the thermodynamic limit required for a reversible process is never fully reached.

The condition usually stated for the derivation is that the patch be much larger than the Planck length and much smaller than the curvature radius, $\ell_P \ll \ell \ll L$. Norton's fluctuation analysis suggests that this is not enough. For the Clausius relation to relate a reliable heat flow to a reliable change of entropy, the entropy change driven through the patch must be much larger than the quantum fluctuations of the horizon entropy, $\delta S \gg 1$ in units of Boltzmann's constant.

Given $\delta S \gg 1$, we now derive the required window of validity for GR as a locally reversible process. Let the patch have transverse size and affine extent of order ℓ, and let $R \sim L^{-2}$. The orders in λ that are dropped (Section 4.4) have relative size of order $(\ell/L)^2$, which gives the upper bound $\ell \ll L$. The entropy change driven by matter is $\delta S \sim \delta A/\ell_P^2 \sim \ell^4/(\ell_P^2 L^2)$, and the condition $\delta S \gg 1$ requires $\ell \gg \sqrt{\ell_P L}$, the geometric mean of the Planck length and the

curvature radius. The window of validity is therefore

$$\sqrt{\ell_P L} \ll \ell \ll L, \tag{6}$$

which is much narrower than $\ell_P \ll \ell \ll L$.

Even this tighter window is almost always satisfied. For a stellar-mass black hole with $L \sim 10\,\mathrm{km}$, $\sqrt{\ell_P L} \sim 10^{-15}\,\mathrm{m}$, so any patch between roughly the size of a proton and a few kilometres qualifies. However, we also have to look at how errors accumulate over time. Near a horizon of radius L, approximately L^2/ℓ^2 patches cover the horizon, and we can treat them as being replaced after a time of order ℓ, so over a time t the horizon involves $N \sim L^2 t/\ell^3$ patches. Assuming the worst case, in which the fluctuations are correlated over time, as the analysis of an evaporating black hole may suggest (Section 5.3), the error grows linearly in N. Errors then become comparable to S after a time of order ℓ^3/ℓ_P^2, which comes well before the Page time, itself of order L^3/ℓ_P^2. This early onset reflects the worst-case assumption. In actual circumstances the correlation will be partial, which shifts the time toward the Page time. In either case the order structure is the same: both times scale as a length cubed over ℓ_P^2, and they coincide as the patch size approaches the horizon radius.

We conclude that, under the local thermodynamic interpretation, GR remains valid as a short-interval law of physics, but problems arise when there are long-term correlations.

5.2 Two ways to keep local reversibility

There are two ways to restore local reversibility.

5.2.1 First option: non-equilibrium corrections from quantum gravity

The first option is that full quantum gravity provides non-equilibrium corrections to the entropy of a reversible process, so that GR, as the same short-interval local thermodynamics, remains valid at any time. Replica wormholes are one possible example [31, 32, 33]. In the gravitational path integral that computes the entropy of Hawking radiation, replica wormholes are additional geometries that connect the different replicas used in the calculation. The physical spacetime is kept as suggested by semiclassical GR, since the wormholes are auxiliary geometries in the calculation rather than part of that spacetime, but the entropy they assign to the radiation is corrected over time, so that it follows the Page curve instead of growing steadily as in Hawking's calculation. Whether a mechanism of this kind also controls the accumulated errors of local horizon thermodynamics is an open question.

5.2.2 Small corrections and unitarity

To preserve the unitarity of the effective description, such corrections from a unitary quantum gravity have to be subtle, and this is in general difficult to achieve. We present a generalization of

Mathur's small corrections theorem [34] in this regard. Let $U_{\mathrm{eq},t}$ be the evolution of the horizon and its surroundings over one step of duration ℓ that keeps every patch in local equilibrium, so that GR holds as short-interval local thermodynamics, and let $U_{\mathrm{eff},t}$ be the evolution corrected by quantum gravity. Let $|\psi_t\rangle$ be the state evolved by U_{eq} up to step t, and suppose that at each step

$$U_{\mathrm{eff},t}|\psi_t\rangle = \alpha_t\, U_{\mathrm{eq},t}|\psi_t\rangle + \sqrt{1-\alpha_t^2}\,|\phi_t\rangle, \qquad \alpha_t^2 = 1 - e^{-S(t)}, \tag{7}$$

with α_t real and non-negative and $S(t)$ large. Taking α_t real is no restriction, since a phase can be absorbed into $U_{\mathrm{eq},t}$. Equation (7) says that the non-equilibrium corrections are small at every step, and their scale $e^{-S(t)}$ follows from what the first option demands. If GR is to hold as short-interval local thermodynamics at every step, the corrections must not change any coarse-grained quantity of a patch in local equilibrium, such as the energy flux, the temperature or the entropy. A thermodynamic description with entropy S leaves about e^S microstates unresolved, and effects confined to this microstate structure are suppressed in amplitude by about $e^{-S/2}$. The eigenstate thermalization hypothesis gives the standard example, where the deviations of individual energy eigenstates from thermal expectation values, and the off-diagonal matrix elements of local observables, are of this size [35, 36, 37]. A correction of larger amplitude would show up in coarse-grained quantities, and so would itself be a departure from local equilibrium. The orthogonal component in Equation (7) therefore has amplitude $\sqrt{1-\alpha_t^2} = e^{-S(t)/2}$. For $S(t)$ one can take the entropy δS driven through a patch or, for the horizon as a whole, the Bekenstein-Hawking entropy. In the first case, the condition $\delta S \gg 1$ behind Equation (6) is what guarantees that $S(t)$ is large.

If $U_{\mathrm{eff},t}$ is unitary, it preserves the norm of the deviation carried over from earlier steps, and the triangle inequality bounds the deviation ϵ_T between the two evolved states after T steps by

$$\epsilon_T \lesssim (T+1)\, e^{-S_{\min}/2}, \qquad S_{\min} \equiv \min_{t \le T} S(t). \tag{8}$$

The number of steps up to the Page time is $T \sim L^3/(\ell_P^2 \ell)$, which grows only polynomially with the entropy, so ϵ_T is still exponentially small at the Page time. If the argument is applied patch by patch with $S(t) = \delta S$, the number of steps T is replaced by the number of patches $N \sim L^2 t/\ell^3$ of Section 5.1, and the conclusion is unchanged as long as δS is larger than $2 \ln N$, which holds well inside the window. A state that deviates this little from the equilibrium-evolved one cannot carry the change of order S in entropy needed to cancel the accumulation of errors found in Section 5.1.

To restore GR as a truly locally reversible process, then, we would likely have to sacrifice the unitarity of the underlying effective quantum theory, even if the corrections are derived from a unitary quantum gravity. Alternatively, the Hilbert space of the effective theory changes over time, and we may attempt to study this by resorting to a topological quantum field theory involving cobordisms [38], having stripped away propagating degrees of freedom, but the

literature on this idea is not yet mature.

5.2.3 Second option: modifying GR

The second option is that GR is modified, with GR then being the local thermodynamic equilibrium limit of a non-equilibrium process or quantum gravity. Such a modification would not be needed in most circumstances. Equation (6), which gives the window of validity, can be satisfied with sufficient margin whenever the curvature radius is far greater than the Planck length, and the accumulation of errors becomes an issue only when the fluctuations of different patches are correlated over long times. A modification of GR would therefore be required only in limited cases, such as a black hole around the Page time, when long-lived correlations have had time to build up, or at late stages of evaporation, when the curvature radius of the horizon approaches the Planck length.

5.3 Why black holes are the hard case

Note that Equation (6) constrains each patch separately. This is sufficient as long as the errors of different patches can be treated as independent, which is reasonable in ordinary circumstances. Vacuum entanglement is short-range and does not persist, and correlations in the local vacuum decay over the scale of a patch. At each moment one can then treat past fluctuations and past entanglement as irrelevant.

Black holes break this assumption. Each Hawking quantum stays entangled with a partner behind the horizon, and the entanglement entropy of the radiation grows steadily up to the Page time [39]. The deviation from local thermodynamic equilibrium accumulates, and the error bound for independent patches no longer bounds the total error. This is the accumulation objection to iterated equilibria from Section 3, now in a gravitational setting. Even though the horizon of a large black hole still has relatively low curvature, a locally reversible process fails there.

6 Conclusion

The paradox of reversible processes has drawn several responses: Norton treats reversible processes as approximations rather than idealizations, Valente and Uffink revive Ehrenfest-Afanassjewa's distinction between quasi-processes and quasi-static processes, epistemic views make reversibility relative to an agent's information, and axiomatic approaches define entropy without reversible processes. The dual time-scale argument often invoked by physicists shows how a reversible process may mathematically be defined, but not how such a process can even ideally be realized in finite time. We suggested that what remains of the paradox is a problem of finite-time approachability: a curve of equilibrium states is consistent as a quasi-process, but nothing guarantees that real processes can approach it in finite time.

Jacobson's derivation of Einstein's equation uses a reversible relation and so commits to local thermodynamic reversibility. Equilibrium is imposed only locally at every point on a family of local Rindler horizons, so that no single horizon has to stay in equilibrium along an extended path. Despite the paradox of reversible processes, the derivation succeeds, because the dynamics that causes trouble, relaxation into equilibrium and dissipation, is stipulated away or pushed to higher order. Once that dynamics is restored, the problem of finite-time approachability returns. Finite horizon patches reach the Clausius equality only up to truncation and fluctuation errors, and a fluctuation criterion in Norton's sense narrows the window of validity to $\sqrt{\ell_P L} \ll \ell \ll L$, where ℓ is the size of the horizon patch, L the local curvature radius and ℓ_P the Planck length.

In ordinary situations the paradox of local reversibility does little harm. Entanglement between neighbouring regions of the vacuum is short-lived, so it does not build up, and the local laws of physics are unaffected by what happened earlier. But black holes are different. They build up long-lived entanglement between the interior and the emitted radiation, which puts pressure on maintaining an approximately locally reversible process, since deviations from local equilibrium accumulate instead of being forgotten. Two responses to the paradox of GR local reversibility are available. Either the (quantum or thermodynamic) fluctuations on the horizon are corrected so that GR remains valid, or GR is modified significantly.

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